WHL 2025

Team Disparity Metrics — Column Reference Guide

This document defines every column produced by the disparity analysis script. Columns are grouped by metric category. Each entry shows the column name, its data type, and the exact formula or logic used to derive it.

Positive disparity values always favour the team. Per-game (_pg) variants normalise totals by total_games for fair cross-team comparison. All float values are rounded to 3 decimal places in the output CSV.

1. Context Columns

These columns provide game volume and win/loss records used to normalise all disparity metrics per game.

Column	Type	Description / Formula
total_games	`Count`	Total matchup records (home_games + away_games)
total_wins	`Count`	Regulation wins as home + regulation wins as away
total_ot_wins	`Count`	Overtime wins as home + overtime wins as away
total_losses	`Count`	All losses (home + away), regardless of OT
win_pct	`Ratio`	(total_wins + 0.5 × total_ot_wins) / total_games
point_pct	`Ratio`	(total_wins × 2 + total_ot_wins) / (total_games × 2) — mirrors points-based standings

2. Goal Disparity

Measures the gap between goals scored and goals conceded. The primary indicator of overall team dominance.

Column	Type	Description / Formula
total_goals	`Count`	All goals scored (home_goals + away_goals)
total_goals_against	`Count`	All goals conceded (home_goals_against + away_goals_against)
goal_disparity	`Integer`	total_goals – total_goals_against | Positive = net scorer, Negative = net conceder
goal_disparity_pg	`Float`	goal_disparity / total_games | Normalised per game for fair comparison across teams

3. Shot Disparity

Captures volume control — how often a team is generating shots vs. defending them. High shot disparity signals sustained offensive zone time.

Column	Type	Description / Formula
total_shots	`Count`	All shots attempted (home_shots + away_shots)
total_shots_against	`Count`	All shots faced (home_shots_against + away_shots_against)
shot_disparity	`Integer`	total_shots – total_shots_against
shot_disparity_pg	`Float`	shot_disparity / total_games

4. Expected Goals (xG) Disparity

xG weights each shot by its probability of scoring based on location, angle, and type. xG disparity is a more reliable predictor of future performance than raw goals.

Column	Type	Description / Formula
total_xg	Float	Cumulative xG generated (home_xg + away_xg)
total_xg_against	Float	Cumulative xG conceded (home_xg_against + away_xg_against)
xg_disparity	Float	total_xg – total_xg_against | Positive = creating better chances than conceding
xg_disparity_pg	Float	xg_disparity / total_games

5. Shot Quality Disparity

Measures the danger level of chances created vs. allowed. A team can win the shot count but still be at a disadvantage if opponent shots come from higher-danger areas.

Column	Type	Description / Formula
shot_quality_for	Float	total_xg / total_shots | Average xG value per shot generated
shot_quality_against	Float	total_xg_against / total_shots_against | Average xG per shot conceded
shot_quality_disparity	Float	shot_quality_for – shot_quality_against | Positive = generating higher-danger chances than conceding

6. Conversion Disparity

Finishing efficiency — how well a team converts shots into goals relative to how well it prevents opponents from doing the same.

Column	Type	Description / Formula
conversion_for	Float	total_goals / total_shots | Goals-per-shot ratio (shooting %)
conversion_against	Float	total_goals_against / total_shots_against | Goals-per-shot conceded (save % inverse)
conversion_disparity	Float	conversion_for – conversion_against | Positive = better finisher than opponent on average

7. Finishing Disparity (Goals vs xG)

Reveals over/under-performance relative to expected output. Large positive values suggest lucky finishing or elite finishing skill; large negative values suggest bad luck or poor finishing.

Column	Type	Description / Formula
finishing_disparity	Float	total_goals – total_xg | Positive = scoring above expected (outperforming xG)
finishing_disparity_pg	Float	finishing_disparity / total_games | Per-game normalised version

8. Penalty Disparity

Reflects discipline imbalance. A negative penalty disparity means a team takes fewer penalties than its opponents — a structural advantage on the power play.

Column	Type	Description / Formula
total_penalties	Count	Total penalties committed by the team
total_penalties_against	Count	Total penalties committed by opponents
penalty_disparity	Integer	total_penalties – total_penalties_against | Negative = more disciplined team

Column	Type	Description / Formula
penalty_disparity_pg	`Float`	penalty_disparity / total_games

9. Home / Away Split

Identifies teams whose performance is heavily dependent on playing at home. A large home_away_split signals a team that struggles to replicate home form on the road.

Column	Type	Description / Formula
home_goal_disparity	`Integer`	home_goals – home_goals_against | Net goal margin in home games only
away_goal_disparity	`Integer`	away_goals – away_goals_against | Net goal margin in away games only
home_away_split	`Integer`	home_goal_disparity – away_goal_disparity | Large value = heavily home-dependent team

10. Composite Disparity Score & Ranking

A single number summarising overall team quality relative to the league. Built from five per-game disparity metrics using z-score standardisation so each contributes equally regardless of scale.

Column	Type	Description / Formula
composite_disparity_score	`Float`	Mean z-score across: goal_disparity_pg, shot_disparity_pg, xg_disparity_pg, shot_quality_disparity, conversion_disparity | z = (x – mean) / std per metric
disparity_rank	`Integer`	Rank 1–32 where 1 = highest composite disparity score (best team)
tier	`Label`	Elite (≥0.75) | Above Average (≥0.25) | Average (≥–0.25) | Below Average (≥–0.75) | Poor (<–0.75)

Quick Reading Guide

You want to know...	Look at...
Which team dominates overall	composite_disparity_score / disparity_rank
Who scores vs. who concedes	goal_disparity_pg
Who controls the puck / shot volume	shot_disparity_pg
Who creates better quality chances	shot_quality_disparity + xg_disparity_pg
Who is lucky vs. sustainable	finishing_disparity_pg (high = lucky, regress expected)
Who is most disciplined	penalty_disparity_pg (negative = fewer penalties taken)
Who relies on home ice	home_away_split (high = inconsistent away from home)
How wins correlate with true quality	Compare win_pct vs composite_disparity_score